Add the directions before the lengths

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Resultant / ES: resultante / PT: resultante

Sum: first tail to final tip.

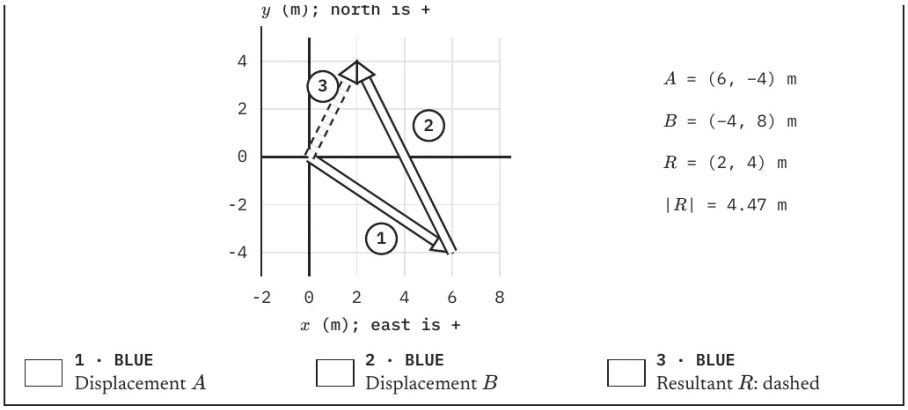
Also on your note page: Component, Magnitude, Quadrant.

B READ THE RELATIONSHIP

Add signed components along each axis

As you read, find: horizontal · vertical · resultant.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Two displacements are A = (6, −4) m and B = (−4, 8) m. Find their sum's magnitude and direction.

Given + x is east + y is north A and B use these axes.

Rule Add x and y . Use the right triangle.

Work R = (2, 4) m | R | = √ 20 m θ = $\tan^{-1}(4/2)$

Answer 4.47 m 63.4° north of east .

E SAVE FOR THE FINAL MEETING

XJ's motion must be tracked in more than one direction. Adding signed components keeps sideways motion in the forecast instead of treating every displacement as motion toward Earth.

Read the physics: Resultant

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

A vector has a size and a direction. Draw the next arrow with its tail at the first arrow's tip. Do not turn the arrow when you move it.

02 HOW IT WORKS

Draw the resultant from the original start to the final tip. To calculate, add horizontal parts together. Add vertical parts separately. The two totals make a right triangle.

03 WHAT TO CHECK

Use the sides to find the length and the signs to choose the direction. State the axis you measured the angle from.

04 TRY THE EXAMPLE FROM YOUR NOTES

Two displacements are A = (6, −4) m and B = (−4, 8) m. Find their sum's magnitude and direction. 4.47 m 63.4° north of east .

Model check: Add signed components along each axis.

05 CONNECT TO XJ

XJ's motion must be tracked in more than one direction. Adding signed components keeps sideways motion in the forecast instead of treating every displacement as motion toward Earth.

SAY IT BEFORE YOU WRITE

Where does the resultant start and end? The two parts are ___ and ___. Together they point ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Resultant / ES: resultante / PT: resultante

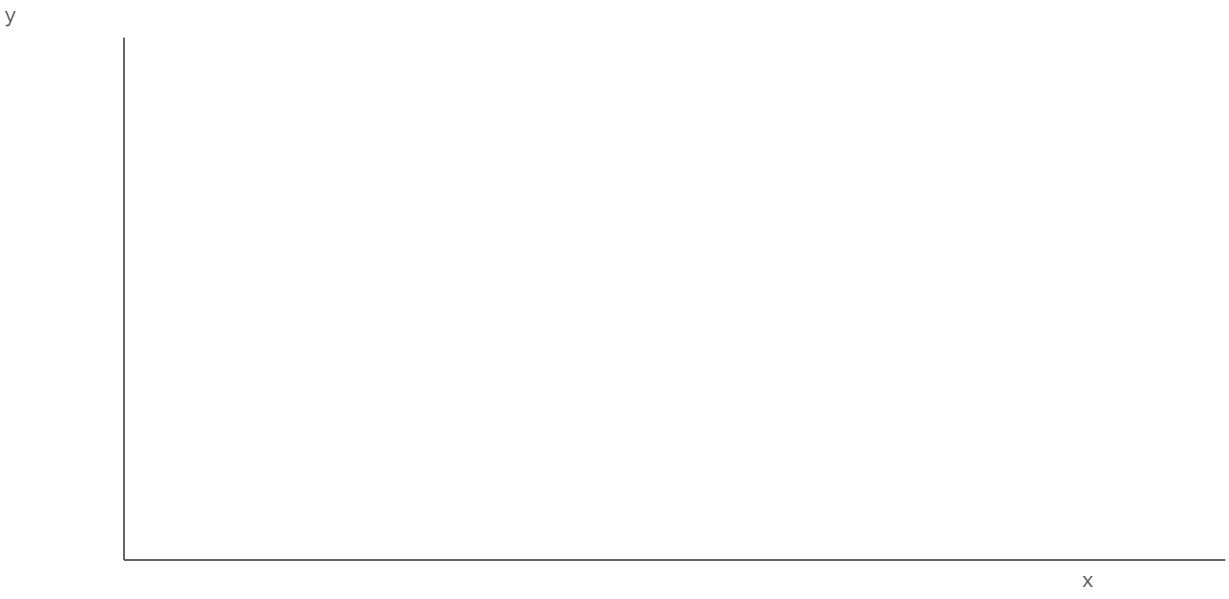
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, resultant means …	If you replace two tip-to-tail arrows with one arrow, where do you put its tail and tip? The single arrow goes from ___ to ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw a right-angle journey; label x, y, and the resultant with its direction.

Labels you can use: horizontal · vertical · resultant.



Tell a partner: The two parts are ___ and ___. Together they point ___.

On assignment: Resultant

SECTION E / BACK / FINAL MEETING ONLY Name _____

A tracker sees a 3 km displacement toward Earth and 4 km sideways. The second telescope needs the whole movement.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Navigation engineer**

On a map, give the camera operator one resultant arrow with a size and direction. Label the two component legs.

☐ **Map presenter**

Deliver a map update. Use one arrow for the displacement and show why "7 km closer to Earth" is the wrong caption.

Word bank: horizontal · vertical · resultant.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."

Editor's check: State the angle reference before explaining your direction.